1. Hill climbing can get stuck due to which of the following phenomenon?
    a. **Local Maxima**
    b. Inadmissible heuristic
    c. **Ridges**
    d. **Plateau**

    Covered in lecture. Inadmissible heuristic is not applicable in case of local search.

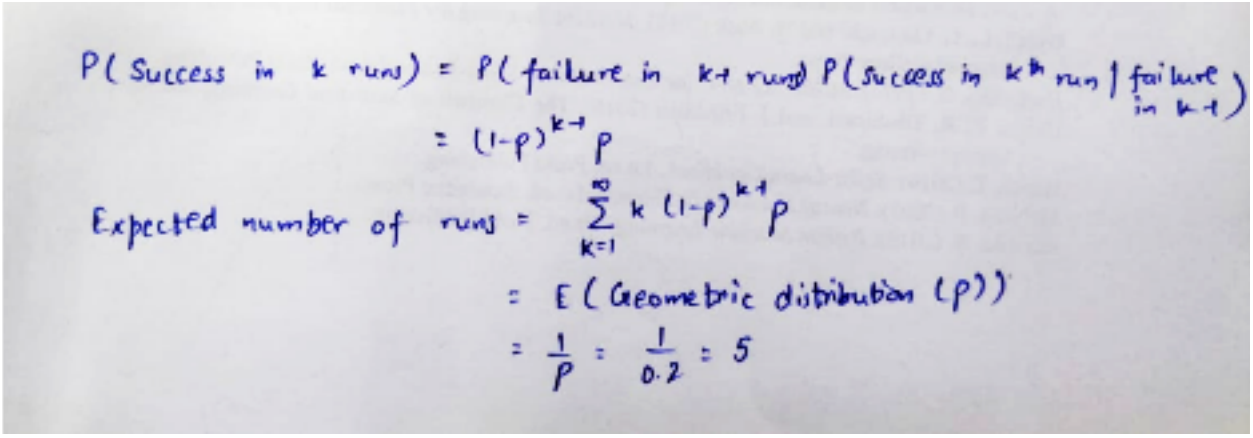
2. Which of the following will eventually find the optimal solution?
    a. **Random Walk**
    b. Greedy Hill Climbing
    c. **Hill Climbing with random restarts**
    d. Local Beam Search

    Only Random Walk and Hill Climbing with random restarts are complete. Greedy Hill Climbing and Local Beam search can get stuck in local optima.

3. What happens to the probability of making a move that leads to a state worse than the current state as we reduce the temperature in simulated annealing?
    a. Increases
    b. **Decreases**
    c. Remains the same

    Probability of taking non-optimal move in simulated annealing is $e^{\Delta E/T}$ and $\Delta E<0$ for non greedy move, thus as temperature reduces $\Delta E/T$ becomes more negative and probability decreases.

4. Suppose a hill-climbing procedure has a probability p=0.2 of getting successful in a particular run. What is the expected number of runs required for hill-climbing with random restarts to be successful?
    Ans. **5**



5. Genetic algorithms are said to jump from one hill to another. Which of the following is responsible for such behavior?
    a. Mutation
    b. Cross over
    c. Fitness function
    d. Natural Selection

    Mutation can only jump to adjacent neighbour. Fitness function is not relevant here. Natural Selection is responsible for the number of offsprings in next generation. Only crossover can do non-local changes.

6. Which of the following pair of algorithms is equivalent?
    a. Tabu Search, Greedy Hill-Climbing with Sideways moves
    b. Simulated Annealing with infinite temperature, Random Walk.
    c. Local Beam Search with infinite beam size (k >> 1), Breadth-First Search.
    d. Genetic Algorithms with unit population size, Random Sampling

    a. Tabu Search maintains a list to store states to avoid making repeated sideways moves whereas Greedy hill climbing with Sideways does not store any such list.
    b. Infinite temperature means the probability of making a random move is 1 which is equivalent to a random walk i.e moving to a random neighboring state.
    c. Breadth first search starts with a single state whereas Local beam search starts with k states. Note that Local Beam search with infinite beam size but single starting state is same as breadth first search.
    d. With unit population size there won't be any crossing over but only mutations in Genetic algorithms which would lead to a random walk not random sampling.

7. Find the no. of steps taken by the Greedy Hill-Climbing Search to go from State A to State B where h(s) = no. of misplaced tiles in state S w.r.t state B.

    Ans: 5

A

3	1	2
4	5	8
6		7

B

	1	2
3	4	5
6	7	8

The following shows the actions of the blank according to the search:
Right - Up - Left - Left - Up. Hence, 5 steps.

8. Which of the following statement(s) is/are correct about Local Beam Search?
    a. It will not find the optimal solution for a beam size of k if it does not find the optimal solution for some beam size > k.
    b. It might find the optimal solution for a beam size of k even if it does not find the optimal solution for some beam size > k.
    c. It has a better chance of finding the optimal solution for higher values of k.
    d. It is the same as Greedy Hill-Climbing with k = 1.

Beam search includes randomness in choosing start states and randomness in choosing the best k. Hence, there is a possibility of finding the solution for a lower k. Because of more exploration, there is a high chance of finding a solution at higher values of k. Trivially for k = 1, it is the same as Greedy Hill-Climbing.

9. What is the direction of steepest descent to find the minimum of the function f(x, y, z) = e^x (xy + 2z) at (x, y, z) = (0, 1, -1).
    a. (2, -1, 0)
    b. (2, 0, -2)
    c. (-1, -1, 0)
    d. (1, 0, -2)

At (x, y, z) = (0, 1, -1).
$\partial f/\partial x = e^x(xy + 2z) + e^x(y) = -1$
$\partial f/\partial y = e^x(x) = 0$
$\partial f/\partial z = 2e^x = 2$
Direction of steepest descent = $(-\partial f/\partial x, -\partial f/\partial y, -\partial f/\partial z) = (1, 0, -2)$

10. Which of the following statement(s) is/are not correct about Tabu Search?
    a. It never visits the same state more than once.
    b. It uses past experience to explore unvisited areas of the search space.
    c. The transition from global to local optimization is marked by increasing the length of tabu list.
    d. It always escapes a local optimum.

Tabu search can visit the same state more than once since the length of the tabu list is finite. It stores the past visited states and uses those to explore new areas. Increasing the length of tabu list allows it to explore more hence it transits to global optimization not local. It has a good chance of escaping local optimum but not always.